

The Sydney-based brain scientist Alan Snyder (who has a Philadelphia accent) made the following discovery. If you inhibit the left hemisphere of a right-handed person (more technically, by directing low-frequency magnetic pulses into the left frontotemporal lobes), you lower his rate of error in reading the above caption. Our propensity to impose meaning and concepts blocks our awareness of the details making up the concept. However, if you zap people's left hemispheres, they become more realistic—they can draw better and with more verisimilitude. Their minds become better at seeing the objects themselves, cleared of theories, narratives, and prejudice.

Why is it hard to avoid interpretation? It is key that, as we saw with the vignette of the Italian scholar, brain functions often operate outside our awareness. You interpret pretty much as you perform other activities deemed automatic and outside your control, like breathing.

What makes nontheorizing *cost* you so much more energy than theorizing? First, there is the impenetrability of the activity. I said that much of it takes place outside of our awareness: if you don't know that you are making the inference, how can you stop yourself unless you stay in a continuous state of alert? And if you need to be continuously on the watch, doesn't that cause fatigue? Try it for an afternoon and see.

A Little More Dopamine

In addition to the story of the left-brain interpreter, we have more physiological evidence of our ingrained pattern seeking, thanks to our growing knowledge of the role of neurotransmitters, the chemicals that are assumed to transport signals between different parts of the brain. It appears that pattern perception increases along with the concentration in the brain of the chemical dopamine. Dopamine also regulates moods and supplies an internal reward system in the brain (not surprisingly, it is found in slightly higher concentrations in the left side of the brains of right-handed persons than on the right side). A higher concentration of dopamine appears to lower skepticism and result in greater vulnerability to pattern detection; an injection of L-dopa, a substance used to treat patients with Parkinson's disease, seems to increase such activity and lowers one's suspension of belief. The person becomes vulnerable to all manner of fads, such as astrology, superstitions, economics, and tarot-card reading.

Actually, as I am writing this, there is news of a pending lawsuit by a patient going after his doctor for more than \$200,000—an amount he allegedly lost while gambling. The patient claims that the treatment of his Parkinson's disease caused him to go on wild betting sprees in casinos. It turns out that one of the side effects of L-dopa is that a small but significant